ESPAÇO VENTORIAL

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1 Groups

A group is a nonempty set equipped with a binary operation $* = G \times G \to G$ satisfying the group axioms below. Commutativity, listed last, is an additional property, not a requirement for every group.

1) Closure: $\forall x, y \in G \to x * y \in G$
2) Associativity: $\forall x, y, z \in G \to x * (y * z) = (x * y) * z$
3) $\exists \xi \in G \ \forall x \in G: \ x * \xi = \xi * x = x.$
4) Inverse: $\forall x \in G \to \exists y \in G \ such \ that \ x * y = y * x = \xi$
5) Commutativity: $\forall x, y \in G \to x * y = y * x$

The identity is denoted by ξ. A set satisfying the first four properties is a group; if it also satisfies commutativity, it is an Abelian group. The identity is one fixed element that works for every group element, not an element chosen separately for each input.

Examples and a Nonexample

1) $(\mathbb{R}, +)$ is an Abelian group.
2) $(\mathbb{R}, -)$ is not a group: ordinary subtraction is not associative and has no two-sided identity.
3) $(\mathbb{R} - 0, \cdot)$ is an Abelian group.
4) $(M_2(\mathbb{R}), +)$, the 2×2 real matrices under addition, is an Abelian group.
5) $(\mathbb{Z}, +)$ is an Abelian group under addition.
6) $(\mathbb{C}, +)$ is an Abelian group under addition.
7) $(\mathbb{p}_n, +)$ is an Abelian group, where $\mathbb{p}_n$ denotes polynomials of degree at most n, including the zero polynomial.

2 Fields

A field is a set with addition and multiplication such that it is an Abelian group under addition, its nonzero elements form an Abelian group under multiplication, and multiplication distributes over addition. The additive and multiplicative identities are distinct.

2.1 Additive Abelian Group under $(+)$

a) Closure: $\forall x, y \in K \rightarrow x + y \in K$
b) Associativity: $\forall x, y, z \in K \rightarrow x + (y + z) = (x + y) + z$
c) Identity: $\forall x \in K \rightarrow \exists 0 \in K \; / \; x \; + 0 = 0 + x = x$
d) Additive inverse: $\forall x \in K \rightarrow \exists y \in K \; / \; x + y = y + x = 0$
e) Commutativity: $\forall x, y \in K \rightarrow x + y = y + x$

2.2 Multiplicative Abelian Group of Nonzero Elements

a) Closure: $\forall x, y \in K \setminus \{0\} \rightarrow x \cdot y \in K \setminus \{0\}$
b) Associativity: $\forall x, y, z \in K \setminus \{0\} \rightarrow x \cdot (y \cdot z) = (x \cdot y) \cdot z$
c) Identity: $\forall x \in K \setminus \{0\} \rightarrow \exists 1 \in K \setminus \{0\} \; / \; x \; \cdot 1 = 1 \cdot x = x$
d) Multiplicative inverse: $\forall x \in K \setminus \{0\} \rightarrow \exists y \neq 0 \in K \setminus \{0\} \; / \; x \cdot y = y \cdot x = 1$
e) Commutativity: $\forall x, y \in K \setminus \{0\} \rightarrow x \cdot y = y \cdot x$
f) Distributivity
g) $\forall x, y, z \in K \rightarrow x \cdot (y + z) = x \cdot y + x \cdot z$

The elements of the field K are called scalars.
Examples
a) The real numbers, $\mathbb{R}$;
b) The complex numbers, $\mathbb{C}$;

c) The rational numbers, $\mathbb{Q} = \left\{\frac{a}{b} \middle| \, a \in \mathbb{Z} \; and \; b \in \mathbb{Z}^*\right\}$.

2.3 Vector Spaces

A vector space E over a field K is a set whose elements form an Abelian group under addition and whose scalar multiplication satisfies the following additional axioms. The two occurrences of E refer to the same underlying set.

a) $\forall \alpha \in K$ and $\forall \boldsymbol{x} \in E \Longrightarrow \alpha \boldsymbol{x} = \boldsymbol{x}\alpha \in E$ (closure under scalar multiplication)

b) $\forall \alpha, \beta \in K$ and $\forall \boldsymbol{x} \in E \rightarrow \alpha(\beta \boldsymbol{x}) = (\alpha\beta)\boldsymbol{x}$

c) $\forall \alpha \in K$ and $\forall \boldsymbol{x}, \boldsymbol{y} \in E \rightarrow \; \alpha(\boldsymbol{x} + \boldsymbol{y}) = \alpha \boldsymbol{x} + \alpha \boldsymbol{y}$

d) $\forall \alpha, \beta \in K$ and $\forall \boldsymbol{x} \in E \rightarrow (\alpha + \beta)\boldsymbol{x} = \alpha \boldsymbol{x} + \beta \boldsymbol{x}$

e) $\exists 1 \in K \; / \; \forall \boldsymbol{x} \in E \rightarrow 1\boldsymbol{x} = \boldsymbol{x}1 = \boldsymbol{x}$

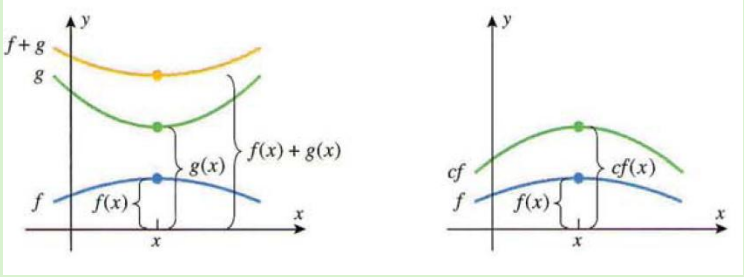
Scalars from K are denoted by Greek letters. Elements of E are called vectors and are written in boldface or with an arrow. Here, 1 denotes the multiplicative identity of the scalar field.

Exercises

1) Show that geometric arrow vectors form a vector space over the real numbers under the usual operations.
2) Show that $E = (x, y) \in \mathbb{R}/y = 2x$ is a vector space over the real numbers.
3) Is $E = (x, y) \in \mathbb{R}/y = 2x + 2$ a vector space under the usual operations? Justify your answer.
4) Show that real polynomials of degree at most three form a vector space over the real numbers. Determine its dimension and give an explicit basis.
5) With the usual operations, $\mathbb{C}$ is a vector space over itself and also over $\mathbb{R}$. Compare the dimensions over the two scalar fields.
6) Consider real-valued functions on a common domain. Functions f and g are equal if and only if $f(x) = g(x)$, $\forall x \in \mathbb{R}$. Define scalar multiplication and addition pointwise:

$$(cf)(x) = cf(x) and (f + g)(x) = f(x) + g(x).$$

The graph of $(f + g)(x)$ is obtained by adding the y coordinates of $f(x)$ and $g(x)$ at each x. Scalar multiplication by c multiplies the value of $f(x)$ by c at every x.



Show that the real-valued functions on $(-\infty, \infty)$, with the operations defined above, form a vector space.

7) Do real square matrices of order two form a vector space under the usual operations?

A notation widely used in quantum mechanics is Dirac's ket notation, $|a\rangle$. It is useful for abstract state vectors and is related to the inner-product structure introduced below. A ket is an abstract vector; a column matrix is its representation after a basis has been chosen.

The dimension of a vector space is the cardinality of a basis. A basis is a linearly independent spanning set, so every vector is a finite linear combination of its elements in the algebraic definition. In an infinite-dimensional Hilbert space, an orthonormal basis instead provides norm-convergent series expansions. For the canonical basis of $\mathbb{R}^3$, e_1, e_2, e_3, a general vector $\boldsymbol{a}$ has the representation

$$a = a_1 e_1 + a_2 e_2 + a_3 e_3 = (a_1, a_2, a_3)^T.$$

The scalar coefficients from R are the components of $\boldsymbol{a}$ along e_1, e_2 and e_3. These components represent $\boldsymbol{a}$ in the chosen basis and change when the basis changes.

A set of vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \cdots, \mathbf{v}_n$ is linearly independent if the only solution of

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 + \cdots + c_n\mathbf{v}_n = 0$$

is $c_1 = c_2 = c_3 = \cdots = c_n = 0$. The vectors themselves are fixed; setting all of them to $\mathbf{v}_1 = \mathbf{v}_2 = \cdots = \mathbf{v}_n = 0$ is not an additional coefficient solution and would violate independence. No vector in $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \cdots, \mathbf{v}_n$ is then a linear combination of the others. A set of n independent vectors in $\mathbb{R}^n$ spans the space, so every vector can be expressed using $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \cdots, \mathbf{v}_n$. A basis is not unique, but coordinates in a fixed basis are unique. Suppose $\mathbf{v}$ has two expansions in the same basis:

$$\mathbf{v} = c_1 e_1 + c_2 e_2 + c_3 e_3 + \cdots + c_n e_n \text{ and } \mathbf{v} = a_1 e_1 + a_2 e_2 + a_3 e_3 + \cdots + a_n e_n.$$

Subtracting gives

$$\mathbf{0} = (c_1 - a_1)e_1 + (c_2 - a_2)e_2 + (c_3 - a_3)e_3 + \cdots + (c_n - a_n)e_n$$

Linear independence implies $c_1 = a_1, c_2 = a_2, \cdots, c_n = a_n$; hence the coordinates are unique.

Exercises

a) Is $(1,0,1), (2,0,0)$ a basis for $\mathbb{R}^3$?

b) Is $(1,1), (0,1)$ a basis for $\mathbb{R}^2$?

c) Is $(1,0), (0,1), (2,1)$ a basis for $\mathbb{R}^2$?

d) Find a basis for the real symmetric matrices of order two.

2.4 Inner Products

An inner product on a real or complex vector space is a scalar-valued map of two vectors:

$$\langle *,* \rangle : E \times E \longrightarrow K$$
$$\langle \boldsymbol{a}, \boldsymbol{b} \rangle \longrightarrow \alpha$$

For all vectors $\forall \boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c} \in E$ and scalars $\forall \alpha, \beta \in K$, it satisfies

a) $\langle \boldsymbol{a}, \boldsymbol{a} \rangle \geq 0$ and $\langle \boldsymbol{a}, \boldsymbol{a} \rangle = 0 \Leftrightarrow \boldsymbol{a} = 0$ — positive definiteness;

b) $\langle \boldsymbol{a}, \boldsymbol{b} \rangle = (\langle \boldsymbol{a}, \boldsymbol{b} \rangle)^* = \langle \boldsymbol{b}, \boldsymbol{a} \rangle^*$ — conjugate symmetry;

c) $\langle \boldsymbol{a}, \alpha \boldsymbol{b} + \beta \boldsymbol{c} \rangle = \alpha \langle \boldsymbol{a}, \boldsymbol{b} \rangle + \beta \langle \boldsymbol{a}, \boldsymbol{c} \rangle$ — linearity in the second argument;

d) $\langle \alpha \boldsymbol{a} + \beta \boldsymbol{b}, \boldsymbol{c} \rangle = \alpha^* \langle \boldsymbol{a}, \boldsymbol{b} \rangle + \beta^* \langle \boldsymbol{a}, \boldsymbol{c} \rangle$ — conjugate linearity in the first argument.

These are the physicists' conventions; some mathematics texts use the opposite linearity convention. An inner-product space is also called a *pre-Hilbert space*. Its norm topology is Hausdorff, but "Hausdorff space" is not a synonym for inner-product space. The usual dot product on $\mathbb{R}^3$ is an inner product. In the canonical basis of $\mathbb{R}^3$: $e_1 = (1,0,0), e_2 = (0,1,0), e_3 = (0,0,1)$,

$$\boldsymbol{a} = a_1 e_1 + a_2 e_2 + a_3 e_3 \text{ and } \boldsymbol{b} = b_1 e_1 + b_2 e_2 + b_3 e_3.$$

Observe that

$$e_i \cdot e_j = |e_i||e_j| cos\theta = 1 \cdot 1 \cdot cos\theta = cos\theta = \begin{cases} 1 & if\ \theta = 0 \\ 0 & if\ \theta = \pi/2 \end{cases}$$

Therefore,

$$\boldsymbol{a}\cdot\boldsymbol{b}=(a_1e_1+a_2e_2+a_3e_3)\cdot(b_1e_1+b_2e_2+b_3e_3)=\sum_{i,j=1}^{3}a_ie_i\cdot b_je_j$$
$$=\sum_{i,j=1}^{3}a_ib_j(e_i\cdot e_j)=\sum_{i,j=1}^{3}a_ib_j\delta_{ij}=\sum_{i=1}^{3}a_ib_i. \quad (1)$$

Equation (1) introduces the Kronecker delta $\boldsymbol{\delta_{ij}}$, defined by

$$\delta_{ij}\equiv\begin{cases}1\ if\ i=j\\0\ if\ i\neq j\end{cases}$$

The vectors e_1,e_2,e_3 form an orthonormal basis for $\mathbb{R}^3$. The construction extends to $\mathbb{R}^n$. For a complex vector space, Dirac notation is convenient. Analogously to e_1,e_2,e_3 in $\mathbb{R}^3$, introduce an orthonormal basis of kets in $\mathbb{C}^n$, denoted by $|i\rangle$, with $i=1,2,3,\cdots,n$. Assume completeness, so any ket $|a\rangle$ can be expanded in $|i\rangle$:

$$|a\rangle=\sum_{i=1}^{n}|i\rangle a_i. \quad (2)$$

In this basis, a vector is determined by the column of coefficients a_i in the expansion of $|a\rangle$ in $|i\rangle$:

$$|a\rangle=\begin{pmatrix}a_1\\a_2\\\vdots\\a_n\end{pmatrix}$$

In finite-dimensional inner-product spaces, the dual consists of linear functionals. The inner product identifies each ket $|a\rangle$ with its bra $\langle a|$ through a conjugate-linear correspondence. In an orthonormal basis, the bra is represented by the conjugate-transposed column. For Hilbert spaces, the corresponding statement concerns the continuous dual and follows from the Riesz representation theorem.

$$\boldsymbol{a}^{\dagger}=(a_1^*,a_2^*,\cdots,a_n^*).$$

The dagger † denotes the conjugate transpose of $\boldsymbol{a}$. The inner product is

$$\langle a||b\rangle=\langle a|b\rangle=\boldsymbol{a}^{\dagger}\boldsymbol{b}=(a_1^*,a_2^*,\cdots,a_n^*)\begin{pmatrix}b_1\\b_2\\\vdots\\b_n\end{pmatrix}=a_1^*b_1+a_2^*b_2+\cdots+a_n^*b_n=\sum_{i=1}^{n}a_i^*b_i$$

In particular,

$$\langle a|a\rangle = \sum_{i=1}^{n} a_i^* a_i = \sum_{i=1}^{n} |a_i|^2$$

is real and nonnegative and is strictly positive for a nonzero vector. It is the squared norm:

$$|\boldsymbol{a}|^2 = \langle a|a\rangle.$$

The dual is a vector space with a dual bra basis $\{\langle i|\}$. Any bra can be expanded as

$$\langle a| = \sum_{i=1}^{n} a_i^* \langle i|. \tag{3}$$

The inner product $\langle a|b\rangle$ becomes

$$\langle a|b\rangle = \left(\sum_{i=1}^{n} a_i^* \langle i|\right)\left(\sum_{j=1}^{n} b_j |j\rangle\right) = \sum_{i,j=1}^{n} a_i^* \langle i|j\rangle b_j.$$

For agreement with Equation (1) and its complex analog, the basis must satisfy

$$\langle i|j\rangle = \delta_{ij}$$

so

$$\langle a|b\rangle = \sum_{i,j=1}^{n} a_i^* \langle i|j\rangle b_j = \sum_{i,j=1}^{n} a_i^* \delta_{ij} b_j = \sum_{i=1}^{n} a_i^* b_i.$$

Thus, ket $|a\rangle$ is represented by a column and bra $\langle a|$ by the row $\boldsymbol{a}^\dagger$. Their inner product is the product of these matrices. To extract component j of $|a\rangle$, multiply from the left by the basis bra $\langle j|$:

$$\langle j|a\rangle = \langle j|\left(\sum_{i=1}^{n} a_i |i\rangle\right) = \sum_{i=1}^{n} \langle j|i\rangle a_i = \sum_{i=1}^{n} \delta_{ij} a_i = a_j \tag{4}$$

and

$$\langle a|j\rangle = \left(\sum_{i=1}^{n} a_i^* \langle i|\right)|j\rangle = \sum_{i=1}^{n} a_i^* \langle i|j\rangle = \sum_{i=1}^{n} a_i^* \delta_{ij} = a_j^*. \tag{5}$$

Substituting Equations (4) and (5) into Equations (2) and (3) gives

$$|a\rangle = \sum_{i=1}^{n} |i\rangle\langle i|a\rangle = \left(\sum_{i=1}^{n} |i\rangle\langle i|\right)|a\rangle. \tag{6}$$

$$\langle a| = \sum_{i=1}^{n} a_i^*\langle i| = \sum_{i=1}^{n} \langle a|i\rangle\langle i| = \langle a|\left(\sum_{i=1}^{n} |i\rangle\langle i|\right). \tag{7}$$

These results imply

$$\sum_{i=1}^{n} |i\rangle\langle i| = \mathbb{1} \tag{8}$$

This is the completeness, or closure, relation. Equation (8) represents the identity operator on the spanned space. For an infinite orthonormal Hilbert basis, the sum converges strongly on vectors. For the three canonical vectors $(1\ 0\ 0), (0\ 1\ 0), (0\ 0\ 1)$,

$$\sum_{i=1}^{n} |i\rangle\langle i| = \begin{pmatrix}1\\0\\0\end{pmatrix}(1\ 0\ 0) + \begin{pmatrix}0\\1\\0\end{pmatrix}(0\ 1\ 0) + \begin{pmatrix}0\\0\\1\end{pmatrix}(0\ 0\ 1) = \begin{pmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{pmatrix}$$

2.5 Norms

For a vector space E over a real or complex field K, a norm on E is a map

$$\|\cdot\|: E \to [0, \infty).$$

satisfying

a) $\|\mathbf{v}\| \geq 0$ and $\|\mathbf{v}\| = 0 \Leftrightarrow \mathbf{v} = 0$, for $\forall \mathbf{v} \in E$;

b) $\|\alpha\mathbf{v}\| = |\alpha|\|\mathbf{v}\|$, for $\forall \mathbf{v} \in E$ and $\alpha \in K$;

c) $\|\mathbf{v} + \mathbf{w}\| \leq \|\mathbf{v}\| + \|\mathbf{w}\|$, for $\forall \mathbf{v}, \mathbf{w} \in E$ (the triangle inequality).

A norm assigns a nonnegative real number, including zero, to each vector; $\mathbb{R}^+$ is understood in this inclusive sense. It induces the distance between two vectors. A vector space equipped with a specified norm is a normed vector space, denoted by $(E, \|\cdot\|)$. A particular vector space definition need not include a norm. An inner product always induces a norm. For geometric real vectors, the dot product is

$$\mathbf{v} \cdot \mathbf{w} = \|\mathbf{v}\|\|\mathbf{w}\| cos\theta.$$

When $\mathbf{v} = \mathbf{w}$, $\mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|\|\mathbf{v}\| cos 0,$, hence $\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}}$. More generally, the norm induced by an inner product is

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}, \quad \forall \mathbf{v} \in E.$$

2.6 Hilbert Spaces

A Hilbert space is a real or complex inner-product space that is complete in its induced norm: every Cauchy sequence converges to an element of that same space. The informal phrase "without holes" refers to this convergence property, not to a separate geometric criterion.

Separability of a Hilbert space $\mathcal{H}$ is an additional property, not part of the general definition. A space is separable if it has a countable dense subset. A set A is at most countable if an injection f from A into $\mathbb{N} = \{0,1,2,3,\cdots\}$ exists. If f is a bijection from A onto $\mathbb{N}$, the set is countably infinite.

For a finite set, cardinality is simply the number of its elements. The cardinality of A is denoted by $|A|$ or $card(A)$.

Examples

a) If $A = \{\text{apple, banana, orange}\}$, then $|A| = 3$.
b) If $B = \{1,2,3,4,5\}$, then $|B| = 5$.
c) If C is the set of weekdays, then $|C| = 7$.
d) If $D = \emptyset$ is the empty set, then $|D| = 0$.

For finite sets, cardinality formalizes counting. Cantor's method compares infinite sets through bijections: two sets have equal cardinality if every element of either set is paired with exactly one element of the other. Comparing a set specifically with the natural numbers determines whether it is countably infinite.

There are different infinite cardinalities. An infinite set is countable if it has the same cardinality as $\mathbb{N} = 1,2,3,4,...$, so its elements can be listed in a sequence. The cardinality of a countably infinite set is $\aleph_0$, read "aleph-null."

Examples

The integers $\mathbb{Z}$, namely $\ldots, -2, -1, 0, 1, 2, \ldots$, are countable. The ordering $0, 1, -1, 2, -2, 3, -3, \ldots$ lists them all, giving $|\mathbb{Z}| = \aleph_0$.

The rational numbers $\mathbb{Q}$ are also countable, despite their density. Fractions can be listed systematically, omitting repetitions; thus $|\mathbb{Q}| = \aleph_0$. An uncountable set cannot be exhausted by a sequence indexed by the natural numbers.

Example

For the real numbers $\mathbb{R}$, Cantor's diagonal argument proves uncountability. Their cardinality, the continuum, is denoted by $\mathfrak{c}$ or $2^{\aleph_0}$. Thus $|\mathbb{R}| > |\mathbb{N}|$, equivalently $\mathfrak{c} > \aleph_0$. A subset A is dense in M when $\bar{A} = M$, where $\bar{A}$ denotes the closure of A, containing every adherence point of A. A point x adheres to A if each neighborhood of x meets A. Equivalently, every ball about x contains a point of A: $B(x,r) \cap A \neq \emptyset$. Interior and boundary points belong to the closure. A boundary point x has every neighborhood meeting both A and its complement, so $B(x,r) \cap A \neq \emptyset$ and

$B(x,r) \cap A^c \neq \emptyset$ for every positive radius. Here, A^c is the complement of A, and closure and complement are taken in the stated ambient space.

Completeness means every Cauchy sequence converges in the space. A sequence $(a_n)_{n\in\mathbb{N}}$ is Cauchy when $\forall\varepsilon > 0,\ \exists n_0 \in \mathbb{N}$ such that $m,n \geq n_0 \Longrightarrow |a_m - a_n| < \varepsilon$. This requires all sufficiently late pairs to be close, with m and n independently large; merely having consecutive differences approach zero is insufficient.

Finite-dimensional examples include $\mathbb{R}^n$ and $\mathbb{C}^n$. An infinite-dimensional example is L, the space of square-integrable complex functions, with functions identified when equal almost everywhere. Its inner product is

$$\langle \varphi(x), \phi(x) \rangle = \int_{-\infty}^{\infty} \varphi^*(x)\phi\,(x)dx \to finite.$$

Square integrability ensures a finite norm and permits normalization of a nonzero wave function. For comparison,

$$\langle e^x, e^{2x} \rangle = \int_{-\infty}^{\infty} e^x e^{2x}\,dx \to diverges.$$

diverges on the real line. These functions do not define normalizable states in that Hilbert space, although other domains or generalized-state descriptions require separate consideration.

2.7 Linear Operators

An operator $\hat{O}\colon E \longrightarrow E$ maps ket $|a\rangle \in E$ to $|b\rangle \in E$, written $\hat{O}|a\rangle = |b\rangle$. There is also a dual right action on bras, $\langle a|\hat{O} = \langle b|$.; it should not be confused with conjugating the ket equation, which introduces the adjoint. After choosing a basis, a ket is represented by a column, a bra by a row, and an operator by a matrix. Thus, the ordinary matrix products $|a\rangle\hat{O}$ and $\hat{O}\langle a|$ have incompatible dimensions. These expressions, $|a\rangle\hat{O}$ and $\hat{O}\langle a|$, are undefined as the indicated ordinary matrix products. Similarly, $|a\rangle|b\rangle$ and $\langle a|\langle b|$ are not inner products. Tensor products or other explicitly defined operations are distinct constructions. For example,

$$|a\rangle|b\rangle = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$$

has no ordinary matrix-product meaning. A linear operator satisfies

$$\hat{O}(\alpha|a\rangle + \beta|b\rangle) = \alpha\hat{O}|a\rangle + \beta\hat{O}|b\rangle, \text{ for } \forall|a\rangle, |b\rangle \in E \text{ and } \forall\alpha,\beta \in K.$$

Example 1. The differentiation operator d/dx is linear:

$$\frac{d}{dx}\big(\alpha f(x)+\beta g(x)\big)=\alpha\frac{df(x)}{dx}+\beta\frac{dg(x)}{dx}.$$

Example 2. Integration with fixed limits and a fixed integration convention is linear:

$$\int[\alpha\, f(x)+\beta\, g(x)]dx=\alpha\int f(x)dx+\beta\int g(x)dx\,.$$

Example 3. The square-root map $\sqrt{\ }$ is generally nonlinear:

$$\sqrt{\alpha f(x)+\beta g(x)}\neq\alpha\sqrt{f(x)}+\beta\sqrt{g(x)}.$$

A finite-dimensional linear operator is determined by its action on a basis. In $\mathbb{R}^2$, its representation is a 2×2 matrix:

$$\hat{\mathcal{O}}=\begin{pmatrix}a & b\\ c & d\end{pmatrix}.$$

For example, let the orthonormal basis be $\{|a\rangle,|b\rangle\}$, with

$$\hat{\mathcal{O}}|a\rangle=|b\rangle \quad \text{and} \quad \hat{\mathcal{O}}|b\rangle=|a\rangle$$

and use the representations $|a\rangle$ and $|b\rangle$ given by

$$|a\rangle=\begin{pmatrix}1\\0\end{pmatrix} \quad \text{and} \quad |b\rangle=\begin{pmatrix}0\\1\end{pmatrix}$$

Then

$$\hat{\mathcal{O}}|a\rangle=|b\rangle\Longrightarrow\begin{pmatrix}a & b\\ c & d\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix}=\begin{pmatrix}0\\1\end{pmatrix}\Longrightarrow\begin{cases}a=0\\ c=1\end{cases}$$

and

$$\hat{\mathcal{O}}|b\rangle=|a\rangle\Longrightarrow\begin{pmatrix}a & b\\ c & d\end{pmatrix}\begin{pmatrix}0\\1\end{pmatrix}=\begin{pmatrix}1\\0\end{pmatrix}\Longrightarrow\begin{cases}b=1\\ d=0\end{cases}$$

The matrix representing $\hat{\mathcal{O}}$ in basis $\{|a\rangle,|b\rangle\}$ is therefore

$$\hat{\mathcal{O}}=\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$$

Now consider a general orthonormal basis $\{|i\rangle\}$. The operator $\hat{\mathcal{O}}$ can be written in bra–ket form. $\hat{\mathcal{O}}|a\rangle=|b\rangle$, the components of $|a\rangle$ and $|b\rangle$ in basis $\{|i\rangle\}$ are

$$|a\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix} \quad \text{and} \quad |b\rangle = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{pmatrix}$$

Applying $\hat{\mathcal{O}}$ to $|a\rangle$ gives

$$\hat{\mathcal{O}}|a\rangle = \begin{pmatrix} \mathcal{O}_{11} & \mathcal{O}_{12} & \cdots & \mathcal{O}_{1n} \\ \mathcal{O}_{21} & \mathcal{O}_{22} & \cdots & \mathcal{O}_{2n} \\ \vdots & \vdots & & \vdots \\ \mathcal{O}_{n1} & \mathcal{O}_{n2} & \cdots & \mathcal{O}_{nn} \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix} = \begin{pmatrix} \sum_j \mathcal{O}_{1j}\alpha_j \\ \sum_j \mathcal{O}_{2j}\alpha_j \\ \vdots \\ \sum_j \mathcal{O}_{nj}\alpha_j \end{pmatrix} = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{pmatrix} \tag{9}$$

Hence,

$$\beta_i = \sum_j \mathcal{O}_{ij}\alpha_j. \tag{10}$$

In basis $\{|i\rangle\}$, the vector $|b\rangle$ is

$$|b\rangle = \sum_i \beta_i |i\rangle \tag{11}$$

Substituting Equation (10) into Equation (11) gives

$$|b\rangle = \hat{\mathcal{O}}|a\rangle = \sum_i \sum_j \mathcal{O}_{ij}\alpha_j \, |i\rangle = \sum_{i,j} \mathcal{O}_{ij}\alpha_j \, |i\rangle \tag{12}$$

Since $|a\rangle = \sum_j \alpha_j \, |j\rangle$, $\alpha_j = \langle j|a\rangle$. Substitute the component α_i into Equation (12):

$$\hat{\mathcal{O}}|a\rangle = \sum_{i,j} \mathcal{O}_{ij}\langle j|a\rangle \, |i\rangle = \sum_{i,j} \mathcal{O}_{ij} \, |i\rangle\langle j|a\rangle = \underbrace{\left(\sum_{i,j} \mathcal{O}_{ij} \, |i\rangle\langle j| \right)}_{\hat{\mathcal{O}}} |a\rangle. \tag{13}$$

The overlap $\langle j|a\rangle$ is a scalar, so it commutes with scalar coefficients and may multiply a vector on either side. Thus, the operator $\hat{\mathcal{O}}$ is

$$\hat{\mathcal{O}} = \sum_{i,j} \mathcal{O}_{ij} \, |i\rangle\langle j|. \tag{14}$$

The dyadic expression $\sum_{i,j} \mathcal{O}_{ij}\,|i\rangle\langle j|$, with coefficients $\mathcal{O}_{ij}$, acts on a vector to produce another vector. In three dimensions,

$$\hat{\mathcal{O}} = \sum_{i,j}^{3} \mathcal{O}_{ij}|i\rangle\langle j|$$

$$= \mathcal{O}_{11}|1\rangle\langle 1| + \mathcal{O}_{12}|1\rangle\langle 2| + \mathcal{O}_{13}|1\rangle\langle 3| + \mathcal{O}_{21}|2\rangle\langle 1| + \mathcal{O}_{22}|2\rangle\langle 2|$$
$$+ \mathcal{O}_{23}|2\rangle\langle 3| + \mathcal{O}_{31}|3\rangle\langle 1| + \mathcal{O}_{32}|3\rangle\langle 2| + \mathcal{O}_{33}|3\rangle\langle 3|$$

$$O = \sum_{i,j=1}^{3} O_{ij}|i\rangle\langle j|, \qquad (|i\rangle\langle j|)_{mn} = \delta_{mi}\delta_{jn}.$$

$$O = \begin{bmatrix} O_{11} & O_{12} & O_{13} \\ O_{21} & O_{22} & O_{23} \\ O_{31} & O_{32} & O_{33} \end{bmatrix}$$

This is the matrix representation of $\hat{\mathcal{O}}$ in the canonical basis $\{|1\rangle, |2\rangle, |3\rangle\}$. Its matrix changes with the basis. Multiplying Equation (14) for $\hat{\mathcal{O}}$ from the left by $\langle m|$ and from the right by $|n\rangle$ gives the matrix elements of $\hat{\mathcal{O}}$:

$$\langle m|\hat{\mathcal{O}}|n\rangle = \sum_{i,j} \mathcal{O}_{ij}\,\langle m|i\rangle\langle j|n\rangle = \sum_{i,j} \mathcal{O}_{ij}\,\delta_{mi}\delta_{jn} = \mathcal{O}_{mn}. \tag{15}$$

Therefore,

$$(\mathbf{O})_{mn} = \langle m|\hat{\mathcal{O}}|n\rangle. \tag{16}$$

Equation (14) makes the action on a ket explicit:

$$O|b\rangle = \sum_{i,j} O_{ij}|i\rangle\langle j|b\rangle = \sum_{i} c_i|i\rangle\,. \tag{17}$$

where

$$c_i = \sum_{j} O_{ij}b_j\,, \quad b_j = \langle j|b\rangle.$$

The right action of $\hat{\mathcal{O}}$ on a bra $\langle a|$ similarly produces a bra:

$$\langle a|O = \sum_{j} d_j\langle j|\,,$$

where

$$d_j = \sum_{i} \quad a_i^* O_{ij}, \quad a_i = \langle i|a\rangle.$$

3 Transpose in Dirac Notation

In the fixed orthonormal basis, the transpose of

$$\hat{\mathcal{O}} = \sum_{i,j} \mathcal{O}_{ij} \, |i\rangle\langle j|$$

is

$$\hat{\mathcal{O}}^T = \sum_{i,j} \mathcal{O}_{j,i} \, |i\rangle\langle j| \text{ or } \sum_{i,j} \mathcal{O}_{ij} \, |j\rangle\langle i| \tag{18}$$

Either transpose the coefficient indices, $\mathcal{O}_{ij} \longrightarrow \mathcal{O}_{ji}$, or interchange the dyad indices, $|i\rangle\langle j| \longrightarrow |j\rangle\langle i|$, but do not do both. Unlike the adjoint, transpose and componentwise conjugation require the chosen basis convention.

4 Complex Conjugation in Dirac Notation

In the chosen basis, componentwise complex conjugation of $\hat{\mathcal{O}}$ gives

$$\hat{\mathcal{O}}^* = \sum_{i,j} \mathcal{O}_{ij}^* \, |i\rangle\langle j|. \tag{19}$$

5 Adjoints and Hermitian Conjugation

The adjoint, or Hermitian conjugate, of $\hat{\mathcal{O}}$ is

$$\hat{\mathcal{O}}^\dagger = \sum_{i,j} \mathcal{O}_{ji}^* \, |i\rangle\langle j| = \sum_{i,j} \mathcal{O}_{ij}^* \, |j\rangle\langle i|. \tag{20}$$

6 Hermitian and Anti-Hermitian Operators

In finite dimensions, $\hat{\mathcal{O}}$ is Hermitian, or self-adjoint, when it equals its conjugate transpose. For unbounded operators in Hilbert space, self-adjointness additionally requires equality of operator and adjoint domains; a formal symmetry calculation alone does not establish that condition.

$$\hat{\mathcal{O}} = \hat{\mathcal{O}}^\dagger.$$

Hermitian examples are the Pauli matrices:

$$\sigma_1 = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$
$$\sigma_2 = \sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$
$$\sigma_3 = \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

An operator is anti-Hermitian when $\hat{\mathcal{O}} = -\hat{\mathcal{O}}^\dagger$. An Anti-Hermitian Example

$$\hat{\mathcal{O}} = \begin{pmatrix} -i & 2+i \\ -2+i & i \end{pmatrix}$$

7 Useful Relations

a) $\left(\hat{\mathrm{A}}\hat{\mathrm{B}}\right)^\dagger = \hat{\mathrm{B}}^\dagger\hat{\mathrm{A}}^\dagger$

b) $\left(\hat{\mathrm{A}}|i\rangle\right)^\dagger = \langle i|\hat{\mathrm{A}}^\dagger$

c) $\left\langle \hat{A}\psi \right| = \langle\psi|\hat{A}^\dagger$

d) $|\phi\rangle + |\psi\rangle = |\gamma\rangle$

e) $|\psi\rangle = \sum_n c_n\,|\psi_n\rangle$

f) $\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*$

g) $|\psi\rangle = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$

h) $\langle\phi| = \begin{pmatrix} d^* & e^* & f^* \end{pmatrix}$

i) $\langle\phi|\psi\rangle = \begin{pmatrix} d^* & e^* & f^* \end{pmatrix} \cdot \begin{pmatrix} a \\ b \\ c \end{pmatrix} = d^*a + e^*b + f^*c$

j) If $\hat{\mathcal{O}} = \hat{\mathcal{O}}^\dagger$ is Hermitian, conjugating the ket equation gives $\left(\hat{\mathcal{O}}|a\rangle = |b\rangle\right)^\dagger \longrightarrow \langle a|\hat{\mathcal{O}} = \langle b|$.

8 Changes of Basis

Consider two orthonormal bases, $\{|i\rangle\}$ and $\{|\alpha\rangle\}$, indexed by Latin and Greek letters. They satisfy $\langle i|j\rangle = \delta_{ij}$ and $\langle\alpha|\beta\rangle = \delta_{\alpha\beta}$ and are complete: $\sum_i |i\rangle\langle i| = \mathbb{1}$ and $\sum_\alpha |\alpha\rangle\langle\alpha| = \mathbb{1}$. Consequently, each vector $|\alpha\rangle$ can be expanded in $\{|i\rangle\}$, namely basis $\{|i\rangle\}$:

$$|\alpha\rangle = \mathbb{1}|\alpha\rangle = \left(\sum_i |i\rangle\langle i|\right)|\alpha\rangle = \sum_i |i\rangle\langle i|\alpha\rangle = \sum_i |i\rangle U_{i\alpha} = \sum_i |i\rangle(\mathbf{U})_{i\alpha} \qquad (21)$$

Define $\langle i|\alpha\rangle \equiv U_{i\alpha} = (\mathbf{U})_{i\alpha}$. Completeness of $\{|\alpha\rangle\}$ also allows each vector in $\{|i\rangle\}$ to be expanded in $\{|\alpha\rangle\}$:

$$|i\rangle = \mathbb{1}|i\rangle = \left(\sum_{\alpha} |\alpha\rangle\langle\alpha|\right)|i\rangle = \sum_{\alpha} |\alpha\rangle\langle\alpha|i\rangle = \sum_{\alpha} |\alpha\rangle\langle i|\alpha\rangle^{*} = \sum_{\alpha} |\alpha\rangle U_{i\alpha}^{*}$$
$$= \sum_{\alpha} |\alpha\rangle(\mathbf{U}^{\dagger})_{\alpha i}. \tag{22}$$

Here, $\langle\alpha|i\rangle = \langle i|\alpha\rangle^{*} = U_{i\alpha}^{*} = (\mathbf{U}^{\dagger})_{\alpha i}$. With the convention for $\mathbf{U}$ in Equation (21), the reversed overlap is a conjugated entry, not an independently relabeled entry: $\langle\alpha|i\rangle \neq U_{\alpha i}$; specifically, $\langle\alpha|i\rangle = U_{i\alpha}^{*}$. Orthonormality gives

$$\delta_{ij} = \langle i|j\rangle = \langle i|\mathbb{1}|j\rangle = \sum_{\alpha} \langle i|\alpha\rangle\,\langle\alpha|j\rangle = \sum_{\alpha} (\mathbf{U})_{i\alpha}\,(\mathbf{U}^{\dagger})_{\alpha j} = (\mathbf{U}\mathbf{U}^{\dagger})_{ij}.$$

Thus, $\mathbb{1} = \mathbf{U}\mathbf{U}^{\dagger}$. Starting instead from $\delta_{\alpha\beta} = \langle\alpha|\beta\rangle$,

$$\delta_{\alpha\beta} = \langle\alpha|\beta\rangle = \langle\alpha|\mathbb{1}|\beta\rangle = \sum_{i} \langle\alpha|i\rangle\,\langle i|\beta\rangle = \sum_{i} \langle i|\alpha\rangle^{*}\,\langle i|\beta\rangle = \sum_{i} U_{i\alpha}^{*} U_{i\beta} = \sum_{i} (\mathbf{U}^{\dagger})_{\alpha i}(\mathbf{U})_{i\beta}$$
$$= (\mathbf{U}^{\dagger}\mathbf{U})_{\alpha\beta}.$$

we obtain $\mathbb{1} = \mathbf{U}^{\dagger}\mathbf{U}$. Therefore, $\mathbf{U}$ is unitary, with $\mathbf{U}^{\dagger} = \mathbf{U}^{-1}$. The bases are related by the unitary transformation in Equation (21), whose entries are the overlaps $\langle i|\alpha\rangle = U_{i\alpha} = (\mathbf{U})_{i\alpha}$ defining $\mathbf{U}$. Let $\mathbf{O}$ represent operator $\hat{\mathcal{O}}$, equivalently $\hat{\mathcal{O}}$, in basis $\{|i\rangle\}$, and let $\mathbf{\Omega}$ be its representation in $\{|\alpha\rangle\}$. Then

$$\hat{\mathcal{O}}|i\rangle = \mathbb{1}\hat{\mathcal{O}}|i\rangle = \left(\sum_{j} |j\rangle\langle j|\right)\hat{\mathcal{O}}|i\rangle = \sum_{j} |j\rangle\langle j|\hat{\mathcal{O}}|i\rangle = \sum_{j} |i\rangle O_{ji} \tag{23}$$

$$\hat{\mathcal{O}}|\alpha\rangle = \mathbb{1}\hat{\mathcal{O}}|\alpha\rangle = \left(\sum_{\beta} |\beta\rangle\langle\beta|\right)\hat{\mathcal{O}}|\alpha\rangle = \sum_{\beta} |\beta\rangle\langle\beta|\hat{\mathcal{O}}|\alpha\rangle = \sum_{\beta} |\beta\rangle\Omega_{\beta\alpha} \tag{24}$$

To relate $\mathbf{\Omega}$ and $\mathbf{O}$, write the matrix element of $\mathbf{\Omega}$ as $\Omega_{\alpha\beta} = \langle\alpha|\hat{\mathcal{O}}|\beta\rangle$ and insert both resolutions of the identity:

$$\Omega_{\alpha\beta} = \langle\alpha|\mathbb{1}\hat{\mathcal{O}}\mathbb{1}|\beta\rangle = \langle\alpha|\sum_{i}|i\rangle\langle i|\,\hat{\mathcal{O}}\,\sum_{j}|j\rangle\langle j|\,|\beta\rangle$$
$$= \sum_{ij} \langle\alpha|i\rangle\langle i|\hat{\mathcal{O}}|j\rangle\langle j|\beta\rangle = \sum_{ij} U_{i\alpha}^{*} O_{ij} U_{j\beta} = \sum_{ij} (\mathbf{U})_{i\alpha}^{*}(\mathbf{O})_{ij}(\mathbf{U})_{j\beta}$$

$$\Omega_{\alpha\beta} = \sum_{ij} (\mathbf{U}^\dagger)_{\alpha i} (\mathbf{O})_{ij} (\mathbf{U})_{j\beta} \tag{25}$$

In matrix notation,

$$\boldsymbol{\Omega} = \mathbf{U}^\dagger \mathbf{O} \mathbf{U} \tag{26}$$

Multiplying Equation (26) from the left by $\mathbf{U}$ and from the right by $\mathbf{U}^\dagger$ gives

$$\mathbf{O} = \mathbf{U} \boldsymbol{\Omega} \mathbf{U}^\dagger \tag{27}$$

Thus, $\mathbf{O}$ and $\boldsymbol{\Omega}$ are unitarily related. In finite dimensions, the spectral theorem guarantees an orthonormal eigenbasis for a Hermitian operator: choose basis $\{|\alpha\rangle\}$ so its matrix is diagonal rather than generally nondiagonal as in $\{|i\rangle\}$:

$$\Omega_{\alpha\beta} = \omega_\alpha \delta_{\alpha\beta} \tag{28}$$

$$\boldsymbol{\Omega} = \mathbf{U}^\dagger \mathbf{O} \mathbf{U} = \boldsymbol{\omega} = \begin{pmatrix} \omega_1 & 0 & \cdots & 0 \\ 0 & \omega_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \omega_n \end{pmatrix}$$

Exercise

1. Find the adjoint of scalar multiplication by $a + bi$.

Solution: if $\hat{O} = a + bi$, with $a, b \in \mathbb{R}$, then $\hat{O}^\dagger = a - bi$.

2. Find the formal adjoint of $\partial/\partial x$ under boundary conditions that remove the boundary term.

$$\langle \phi | D\varphi \rangle = \int_{-\infty}^{\infty} \phi^*(x) \varphi'(x)\, dx$$

$$= [\phi^*(x)\varphi(x)]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} (\phi'(x))^* \varphi(x)\, dx$$

$$= -\langle D\phi | \varphi \rangle, \quad D = \frac{d}{dx}, \quad D^\dagger = -D \text{ formally.}$$

Integration by parts gives the second equality. The boundary term $[\phi^*(x)\varphi(x)]_{-\infty}^{\infty}$ vanishes for suitable functions $\phi(x)$ and $\varphi(x)$, such as smooth compactly supported functions. Thus, the formal adjoint of $\partial/\partial x$ is $-\partial/\partial x$. A Hilbert-space adjoint requires explicit operator domains.

3. Find the adjoint of

$$\hat{O} = \begin{pmatrix} 1 & 0 \\ -i & 1 \end{pmatrix}.$$

$$O^\dagger = \begin{bmatrix} 1 & i \\ 0 & 1 \end{bmatrix}$$

4. Show that the identity is Hermitian.

$$\hat{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \hat{I}^\dagger.$$

5. Show that the following operator is anti-Hermitian.

$$\hat{\mathcal{O}} = \begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix}$$

$$\hat{\mathcal{O}} = \begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix} \Rightarrow \hat{\mathcal{O}}^\dagger = \begin{pmatrix} -i & 0 \\ 0 & -i \end{pmatrix} = -\begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix} = -\hat{\mathcal{O}}.$$

6. Show that momentum $\hat{p}_x$ is formally Hermitian under appropriate boundary conditions.

$$\hat{p}_x = -i\hbar \frac{\partial}{\partial x}$$

Let $\varphi(x)$ and $\phi(x)$ belong to a common suitable domain:

$$\left\langle -i\hbar \frac{\partial}{\partial x}\varphi(x) \middle| \phi(x) \right\rangle = \left\langle \varphi(x) \middle| \left(-i\hbar \frac{\partial}{\partial x}\right)^\dagger \phi(x) \right\rangle = \left\langle \varphi(x) \middle| \left(\frac{\partial}{\partial x}\right)^\dagger (-i\hbar)^\dagger \phi(x) \right\rangle = \left\langle \varphi(x) \middle| \left(\frac{\partial}{\partial x}\right)^\dagger i\hbar\phi(x) \right\rangle$$

$$= \left\langle \varphi(x) \middle| -\frac{\partial}{\partial x} i\hbar\phi(x) \right\rangle = \left\langle \varphi(x) \middle| -i\hbar \frac{\partial}{\partial x}\phi(x) \right\rangle$$

Thus, $\hat{p}^\dagger = \hat{p}$ formally, and $\hat{p}$ is symmetric on a suitable common domain. Self-adjointness requires the additional domain analysis.

7. Show that $\partial^2/\partial x^2$ is formally Hermitian under suitable boundary conditions.

$$\left\langle \frac{\partial^2}{\partial x^2}\varphi(x) \middle| \phi(x) \right\rangle = \left\langle \varphi(x) \middle| \left(\frac{\partial^2}{\partial x^2}\right)^\dagger \phi(x) \right\rangle = \left\langle \varphi(x) \middle| \left(\frac{\partial}{\partial x}\frac{\partial}{\partial x}\right)^\dagger \phi(x) \right\rangle = \left\langle \varphi(x) \middle| -\frac{\partial}{\partial x}\left(-\frac{\partial}{\partial x}\right)\phi(x) \right\rangle$$

$$= \left\langle \varphi(x) \middle| \frac{\partial^2}{\partial x^2}\phi(x) \right\rangle.$$

8. Show that the Schrödinger operator $\hat{\mathcal{H}}$ is formally Hermitian for a real potential and suitable boundary conditions.

$$\langle \hat{\mathcal{H}}\varphi(x) | \phi(x) \rangle = \left\langle \left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + \hat{V}\right]\varphi(x) \middle| \phi(x) \right\rangle = \left\langle \varphi(x) \middle| \left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + \hat{V}\right]^\dagger \phi(x) \right\rangle$$

$$= \left\langle \varphi(x) \middle| \left[\left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\right)^\dagger + (\hat{V})^\dagger\right]\phi(x) \right\rangle = \left\langle \varphi(x) \middle| \left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + \hat{V}\right]\phi(x) \right\rangle = \langle \varphi(x) | \hat{\mathcal{H}}\phi(x) \rangle.$$

Therefore, $\hat{\mathcal{H}} = \hat{\mathcal{H}}^\dagger$ formally on the stated domain

9 Eigenvalues and Eigenvectors

An operator sends a vector to another vector: $\hat{\mathcal{O}}: E \longrightarrow E$, with the corresponding adjoint action on the dual $\hat{\mathcal{O}}: E^* \longrightarrow E^*$. In particular,

$$\begin{aligned} \hat{\mathcal{O}}|a\rangle &= |b\rangle \\ \langle a|\hat{\mathcal{O}}^\dagger &= \langle b| \end{aligned} \tag{29}$$

For an eigenvector, the action reduces to scalar multiplication, which may include a complex phase as well as a change in magnitude:

$$\hat{\mathcal{O}}|\alpha\rangle = \omega_\alpha|\alpha\rangle \tag{30}$$

$$\langle\alpha|\hat{\mathcal{O}}^\dagger = \omega_\alpha^*\langle\alpha| \tag{31}$$

The constants ω_α and ω_α^* are the eigenvalue and its conjugate; $|\alpha\rangle$ is a nonzero right eigenvector, and $\langle\alpha|$ satisfies the corresponding adjoint bra equation.

10 Expectation Values of Hermitian Operators Are Real

For a normalized vector $\varphi(x)$, the expectation of $\hat{\mathcal{O}}$ is $\langle\varphi(x)|\hat{\mathcal{O}}|\varphi(x)\rangle$. Since $\hat{\mathcal{O}}$ satisfies $\hat{\mathcal{O}} = \hat{\mathcal{O}}^\dagger$,

$$\langle\varphi|O|\varphi\rangle^* = \langle\varphi|O^\dagger|\varphi\rangle = \langle\varphi|O|\varphi\rangle.$$

the expectation is real. For an unnormalized nonzero vector, divide by its squared norm.

11 Eigenvalues of Hermitian Operators Are Real

Let $\hat{\mathcal{O}}$ satisfy $\hat{\mathcal{O}} = \hat{\mathcal{O}}^\dagger$. Multiply Equation (30) by bra $\langle\alpha|$ and Equation (31) by ket $|\alpha\rangle$, then subtract:

$$\begin{aligned} \langle\alpha|\hat{\mathcal{O}}|\alpha\rangle &= \omega_\alpha\langle\alpha|\alpha\rangle \\ - \\ \underline{\langle\alpha|\hat{\mathcal{O}}^\dagger|\alpha\rangle &= \omega_\alpha^*\langle\alpha|\alpha\rangle} \\ 0 &= (\omega_\alpha - \omega_\alpha^*)\langle\alpha|\alpha\rangle \end{aligned} \tag{32}$$

Positive definiteness gives $\langle\alpha|\alpha\rangle > 0$ for a nonzero eigenvector, so

$$\omega_\alpha - \omega_\alpha^* = 0 \Longrightarrow \omega_\alpha = \omega_\alpha^*$$

12 Eigenvectors with Distinct Eigenvalues Are Orthogonal

Let $|\alpha\rangle$ and $|\beta\rangle$ be eigenvectors of a Hermitian operator $\hat{\mathcal{O}}$, with $\hat{\mathcal{O}} = \hat{\mathcal{O}}^\dagger$:

$$\hat{\mathcal{O}}|\alpha\rangle = \omega_\alpha|\alpha\rangle \tag{33}$$

$$\langle\beta|\hat{\mathcal{O}}^\dagger = \omega_\beta^*\langle\beta| \tag{34}$$

Multiply Equation (33) from the left by $\langle\beta|$ and Equation (34) from the right by $|\alpha\rangle$, then subtract:

$$\begin{array}{rl} & \langle\beta|\hat{\mathcal{O}}|\alpha\rangle = \omega_\alpha\langle\beta|\alpha\rangle \\ - & \\ & \langle\beta|\hat{\mathcal{O}}^\dagger|\alpha\rangle = \omega_\beta^*\langle\beta|\alpha\rangle \\ \hline 0 & = (\omega_\alpha - \omega_\beta^*)\langle\beta|\alpha\rangle \end{array}$$

Equivalently,

$$(\omega_\alpha - \omega_\beta)\langle\beta|\alpha\rangle = 0. \tag{35}$$

Here, $\omega_\beta^* = \omega_\beta$, because Hermitian eigenvalues are real. Equation (35) implies $\langle\beta|\alpha\rangle = 0$ when $\omega_\alpha \neq \omega_\beta$. Within a degenerate eigenspace, eigenvectors need not initially be orthogonal, but you can choose an orthonormal basis, for example by Gram–Schmidt orthogonalization.

13 The Dirac Delta Distribution

The Dirac delta $\delta(x)$ is a distribution rather than an ordinary function. The translated delta $\delta(x - x_0)$ is supported at its center and has unit total mass. The notation below describes its support, not an infinite pointwise value:

$$\begin{gathered} \operatorname{supp}\delta(x - x_0) = \{x_0\}. \\ \text{Its normalization is} \\ \int_{-\infty}^{\infty} \delta(x - x_0)dx = 1 \end{gathered} \tag{36}$$

The action of $\delta(x - x_0)$ is well-defined distributionally. For a smooth test function $f(x)$, the delta samples its value at x_0:

$$\int_{-\infty}^{\infty} f(x)\delta(x - x_0)dx = f(x_0).$$

Thus, the integral selects the value of $f(x)$, namely $f(x)$, at $x = x_0$.